

Sample PDF for Mini RAG Chatbot Assignment

This PDF is created as a sample dataset for a Retrieval-Augmented Generation (RAG) chatbot project. The chatbot can ingest this document, split it into chunks, create embeddings, store them in a vector database, and answer user questions based on the content.

What is RAG?

Retrieval-Augmented Generation (RAG) is an AI architecture that combines information retrieval with large language models. Instead of relying only on pretrained knowledge, the chatbot retrieves relevant documents from an external knowledge base and uses them to generate accurate responses.

Components of a RAG System

A basic RAG system contains document loading, text chunking, embedding generation, vector storage, retrieval, prompt construction, and response generation using an LLM.

Embeddings

Embeddings are numerical vector representations of text. Similar texts have similar embeddings. Popular embedding models include OpenAI embeddings and Sentence Transformers.

Vector Databases

Vector databases store embeddings and perform similarity search. Common vector databases include FAISS, ChromaDB, Pinecone, and Weaviate.

Chunking

Chunking divides large documents into smaller sections before embedding. Smaller chunks improve retrieval quality and reduce hallucination.

Prompt Engineering

Prompt engineering is the process of designing instructions for language models. In RAG systems, retrieved context is inserted into prompts before generating answers.

Applications of RAG

RAG is used in customer support chatbots, internal company knowledge assistants, educational tools, research assistants, legal document search, and healthcare information systems.

Common Technologies Used in RAG

Technology	Purpose
LangChain	Framework for building RAG pipelines
ChromaDB	Vector database for embeddings
OpenAI API	Embedding and LLM generation
Streamlit	Frontend web interface
PyPDFLoader	Loads PDF documents

Sample FAQ Section

- Q:** What is the role of retrieval in RAG?

A: Retrieval fetches relevant chunks from the vector database based on semantic similarity.
- Q:** Why use chunking?

A: Chunking helps the retriever find more precise information and improves answer accuracy.
- Q:** Can RAG reduce hallucinations?

A: Yes. Since responses are grounded in retrieved documents, hallucinations are reduced.
- Q:** What is semantic search?

A: Semantic search finds information based on meaning instead of exact keyword matching.